

Malak Hasan

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PROFESSIONAL SUMMARY

Computer Engineering graduate specializing in Artificial Intelligence, Machine Learning, and Computer Vision, with hands on experience in deep learning model development, computer vision systems, and cross platform mobile application engineering. Proven ability to design and ship end to end AI solutions, from data preprocessing and model training to backend integration and deployment, across diverse real world applications. Passionate about building intelligent systems that solve meaningful problems.

EDUCATION

Lebanese International University

Bachelor of Science in Computer Engineering | 2022 - 2026

Relevant Coursework: Data Structures & Algorithms, Operating Systems, Machine Learning, Scripting Languages, Computer Vision, Database Systems

EXPERIENCE

Flutter Mobile Development Intern | *Coding and Create* | Jul 2025 – Aug 2025

- Built and delivered cross platform mobile UI screens using Flutter and Dart for iOS and Android, covering multiple app features across a 2 month internship.
- Implemented responsive UI design patterns that resolved layout inconsistencies across device sizes, ensuring a consistent experience on both iOS and Android.
- Optimized app performance by identifying and resolving loading bottlenecks, reducing startup delays and improving runtime responsiveness.
- Collaborated in an Agile team environment to support feature implementation, unit testing, and debugging cycles.

AI BootCamp Trainee | *SprintOS & GDG Coast Lebanon* | Jul 2025

- Completed an intensive 2-week AI bootcamp covering machine learning and deep learning fundamentals hosted by SprintOS and GDG Coast Lebanon.
- Built a stomach cancer detection system as a team project, contributing across data preprocessing, CNN model architecture, training, and evaluation using TensorFlow and Keras.

PROJECTS

Stomach Cancer Detection System

Tech Stack: *Python, TensorFlow, Keras, OpenCV, NumPy*

- Designed and implemented a deep learning based medical image classification system for automated stomach cancer detection using Convolutional Neural Networks (CNNs).
- Performed data preprocessing, augmentation, and normalization on medical imaging datasets to improve model generalization and reduce overfitting.
- Trained and evaluated classification models using TensorFlow and Keras, achieving reliable diagnostic accuracy on held out test datasets.
- Integrated the trained model into an end to end inference pipeline; collaborated with the team on validation, testing, and final presentation.

Role: Data Preprocessing · Model Training & Evaluation · Workflow Integration

SPARK – Virtual Lab Assistant

Tech Stack: *Python, Flask, Google Gemini 2.5 Flash, OpenCV, MediaPipe, JavaScript, HTML/CSS*

- Designed and developed an AI powered chemistry lab safety system for Lebanese schools, integrating real time camera based monitoring for PPE compliance, chemical identification via QR codes, and hazardous pour gesture detection using MediaPipe.

- Built a ChemAI conversational assistant powered by Google Gemini, enabling students to ask chemistry questions, analyze chemical reactions from live camera frames, and check chemical compatibility in real time.
- Implemented a virtual lab environment with hand tracking (MediaPipe Hands) for interactive chemistry experiments, and a full body lab avatar using MediaPipe Pose and FaceMesh for immersive user engagement.
- Architected a single page Flask web application with threaded camera processing, voice input via Web Speech API, and a modular backend serving all features through a unified real time pipeline.

Role: System Architecture · AI Integration · Computer Vision · Frontend and Backend Development

TECHNICAL SKILLS

Languages: Python, Dart

Frameworks & Libraries: TensorFlow, Keras, OpenCV, Flask, Flutter, NumPy

AI / ML: Machine Learning, Deep Learning, Computer Vision, Convolutional Neural Networks (CNN), Image Classification, Model Training & Evaluation, Data Preprocessing, Transfer Learning

Mobile Development: Cross Platform UI, Responsive Design, Performance Optimization

Tools & Practices: Git, GitHub, Technical Documentation, Debugging, Unit Testing, Agile Collaboration

CERTIFICATIONS

Deep Learning Track – AI Summer Sprint Bootcamp | SprintOS & GDG Coast Lebanon | 2025

AI & Technology Bootcamp – Techsplore 2025 | All Girls Code | 2025